CURRICULUM COMPANION : TECHNICAL LEARNING MODULES & FREE COURSES

Week 1 Learning Materials : Foundations & Data Engineering

Core Concepts, Defensive Ingestion, Leak-Free Pipelines, 100% Free Courses & Documentation

TARGET AUDIENCE	ESTIMATED READING TIME	COURSE COST	PRIMARY OBJECTIVE
ML Interns (Trial Phase)	40-50 Minutes	100% Free (Zero Paywalls)	Leak-Free Pipeline Discipline

01 Virtual Environment Isolation & Atomic Git Hygiene

MODULE 1

Isolated environments prevent package conflicts between client projects. Pin exact versions in **`requirements.txt`** to ensure zero production drift. Commit code atomically following Conventional Commits format (`feat:`, `fix:`, `test:`, `docs:`). Never commit large raw data or virtual environment artifacts to version control.

```
# Virtualenv initialization and dependency management
$ python -m venv .venv && source .venv/bin/activate # On Windows: .\.venv\Scripts\Activate.ps1
$ pip install -r requirements.txt && pytest -v && ruff check .
```

02 Defensive Data Ingestion & Health Diagnostics

MODULE 2

Real-world telemetry data arrives with missing timestamps, corrupt rows, and unexpected encodings. Your loader must fail fast on invalid files and compute memory diagnostics before running training:

DEFENSIVE INGESTION RULEBOOK

1. Defensive Type Casting: Force string columns to strip whitespace; replace empty strings and "nan" tokens with explicit "UNKNOWN" categories.
2. Skew-Aware Imputation: Never use mean imputation on right-skewed operational metrics (such as session duration). Always use median values to resist extreme outliers.
3. Memory Footprint: Downcast 64-bit integers to 32-bit where allowable and convert low-cardinality strings to categorical dtypes to save 60-80% RAM.

03 Strict Data Leakage Safeguards & Stratification

MODULE 3

Data Leakage occurs when information from outside the training dataset is used to create or tune the model. A common mistake is fitting scalers or imputers on the whole dataset before splitting.

THREE UNBREAKABLE RULES OF LEAK-FREE SPLITTING

- ✓ **Split Before Transforming:** Partition data into train (70%), validation (15%), and test (15%) splits before calculating ANY mean, median, IQR, or scaler parameters.
- ✓ **Index Disjointness:** Formally verify with sets that `set(train_idx).isdisjoint(test_idx)` evaluates to True.
- ✓ **Stratified Sampling:** In imbalanced datasets (e.g. 15% fraud), random splitting can yield unrepresentative test subsets. Always stratify across the target label.

04 High-Performance Columnar Storage : Parquet vs CSV

MODULE 4

While CSV stores data row-by-row as uncompressed text, Apache Parquet uses columnar encoding with embedded schema metadata, snappy compression, and column pruning capabilities:

FEATURE	CSV (COMMA-SEPARATED)	APACHE PARQUET (SNAPPY)
Storage Format	Row-oriented plaintext	Column-oriented binary
Disk Footprint	Large (e.g. 2.63 MB raw benchmark)	Compact (approx 0.7 - 0.9 MB with Snappy)
Schema Preservation	None; dtypes re-inferred on each load	Strict embedded schema & column dtypes
Query Performance	Must scan entire file sequentially	Reads only requested columns (column projection)

05 Statistical Hypothesis Testing Frameworks

MODULE 5

Never rely solely on visual impressions from charts. Ground your findings in statistical hypothesis testing:

TEST A Chi-Square Test of Independence (χ^2)

Categorical vs Categorical

Evaluates whether two categorical variables (e.g. Device OS vs Fraud Flag) are statistically independent.

```
# Null Hypothesis (H0): Device OS and Abuse Rate are independent.
>>> chi2, p_val, dof, expected = scipy.stats.chi2_contingency(contingency_table)
# Decision: If p_val < 0.05, reject H0 (statistically significant association exists).
```

TEST B Mann-Whitney U Test (Wilcoxon Rank-Sum)

Continuous vs Binary Label

Non-parametric test comparing the distributions of two independent groups when normality cannot be assumed (e.g. skewed session durations).

```
# Null Hypothesis (H0): Normal and Fraudulent sessions share identical distributions.
>>> stat, p_val = scipy.stats.mannwhitneyu(benign_durations, fraud_durations)
```

06 Unit Testing Data Pipelines & Code Quality

MODULE 6

- ☐ **Empty File Handling:** Verify `load_data` raises `ValueError` on 0-byte files.
- ☐ **Zero Data Leakage:** Assert train, val, and test index sets are completely disjoint.
- ☐ **Split Ratio Accuracy:** Verify sum of split lengths equals exactly total row count.
- ☐ **Documentation & Typing:** Every function has complete Google docstrings and type annotations.

WEEKEND 1-ON-1 MENTOR PROBE PREPARATION

During your weekend feedback session, your ML Lead will review your PR and ask: (1) "Why did you choose median over mean imputation for telemetry metrics?", (2) "Prove that your preprocessing pipeline introduces zero data leakage," and (3) "What did your Chi-Square contingency table reveal regarding device abuse?" Be prepared to walk through your code and charts directly.

07 Curated 100% Free Courses & Interactive Practice Labs

ZERO PAYWALLS

All courses listed below are **100% free** with no paywalls, subscriptions, or credit card requirements. They provide interactive coding environments and video walkthroughs directly aligned with Week 1 deliverables:

COURSE & PROVIDER	PLATFORM & FORMAT	DURATION	CORE SKILLS & DIRECT APPLICATION
Data Cleaning Kaggle Learn	Interactive Jupyter Labs	4 Hours	Handling missing values, parsing dates, encoding cleanup, and inconsistent data entry resolution (Task 1.2).
Pandas Kaggle Learn	Interactive Coding Labs	4 Hours	Vectorized operations, multi-index sorting, grouping, summary metrics, and defensive data transformations (Task 1.1).
Data Visualization Kaggle Learn	Interactive Jupyter Labs	4 Hours	Seaborn distribution plots, correlation heatmaps, boxplots, and categorical bar charts for Task 1.3 insights.
Missing Semester of CS MIT OpenCourseWare	Lectures + Exercises	Self-Paced	Shell scripting, virtual environments, Git version control, command-line data wrangling, and debugging hygiene.
CS50P: Testing & Packaging Harvard University	Video + Sandbox Labs	3-4 Hours	Writing automated unit tests with `pytest`, raising defensive exceptions, and modular package structure.
StatQuest: Hypothesis Tests Josh Starmer (YouTube)	Illustrated Video Series	2-3 Hours	Intuitive explanations of Chi-Square test of independence, p-values, Mann-Whitney U test, and IQR outlier boundaries.

08 Official Documentation & Technical Cheat Sheets

PRIMARY REFERENCES

Bookmark these primary documentation hubs for immediate reference when designing your utilities and pipelines:

PRIMARY DOCUMENTATION INDEX

1. Scikit-Learn Common Pitfalls: `scikit-learn.org/stable/common_pitfalls.html` (Official guide on preventing data leakage and test-set contamination).
2. Apache PyArrow Documentation: `arrow.apache.org/docs/python/parquet.html` (Columnar format, Snappy compression, and zero-copy memory reads).
3. Pandas Working with Missing Data: `pandas.pydata.org/docs/user_guide/missing_data.html` (Nullable dtypes, boolean masks, and fillna patterns).
4. Pytest Getting Started & Fixtures: `docs.pytest.org/en/stable/getting-started.html` (Test organization, fixtures, and assertion reporting).
5. Astral Ruff Documentation: `docs.astral.sh/ruff/` (Modern high-speed Python linting, rule selection, and automated fixes).

09 Recommended 5-Day Study & Execution Plan

WORKFLOW SCHEDULE

- ☐ **Day 1 (Scaffold):** Watch MIT Missing Semester (Git/venv) & clone starter repository.
- ☐ **Day 2 (Utilities):** Review CS50P Pytest module & build `src/utils.py` with tests.
- ☐ **Day 3 (Cleaning):** Complete Kaggle Data Cleaning & generate `clean.parquet`.
- ☐ **Day 4 (EDA):** Watch StatQuest Chi-Square video & complete statistical tests in notebook.
- ☐ **Day 5 (Review):** Author Executive Insights Memo & push final PR before Friday 11:59 PM.
- ☐ **Weekend:** Review Mentor Probe Cheatsheet and attend scheduled 1-on-1 feedback session.